Md Al Amin

ahmedalamin@chosun.ac.kr Google Scholar GitHub

Education

Feb 2018 – Dec 2022 **Bachelor of Computer Science and Engineering,** Prime University, Bangladesh.
Thesis Title: *Using Statistical Bioinformatics to Reveal the Biological Mechanisms Linking Smoking and Type 2 Diabetes through Transcriptomic and GWAS Analysis.*
CGPA: *3.81 out of 4.0*
Total Credits: *157.5*

Mar 2024 – Feb 2026 **M.Sc. in Computer Engineering,** Chosun University, South Korea.
Thesis Title: *Localized Receptive Field-wise Image Feature Learning.*
Position: Research Assistant in the Computer Vision Lab.
CGPA: *4.06 / 4.50*
Total Credits: *24*

Interests

Artificial Intelligence

- Machine Learning & Deep Learning
- Image Processing & Computer Vision
- Explainable AI
- Human-AI Interaction
- Quantum Optimization
- Bioinformatics

Feature Engineering

- Tendency to use any type of data in machine learning.
- Finding ways to prepare data for use in machine learning.

Projects

- **Computer Vision:** Steering angle prediction by CNN based on images of lane lines.
- **Computer Vision:** Fine-grained image classification using convolutional neural networks in PyTorch.
- **Computer Vision:** Fake Fingerprint Detection using convolutional neural networks in PyTorch.
- **Generative Adversarial Networks:** Created number images using GANs on the MNIST dataset.
- **Natural Language Processing:** 20 Newsgroups Classification using Attention-based BiLSTM.
- **Natural Language Processing:** Applied machine learning in finding pair question similarity in Quora.

Research Publications (Total Publications: 19, Total Citations: 412)

Journal Articles

1. N. Reza, M. Al Amin, and H. Y. Jung, "Mcl: Meta convolution layer," *IEEE Access*, 2026, Publication Status: Under Review, Journal: Q1, Impact Factor: 3.6.
2. M. Al Amin and H. Y. Jung, "Fine grained visual classification: A survey of methods, architectures, and emerging trends," *IEEE Access*, 2026, Journal: Q1, Impact Factor: 3.6. DOI: https://doi.org/10.1109/ACCESS.2026.3681899.

3. M. Al Amin, N. Reza, and H. Y. Jung, "Localized feature enhancement via patch-wise learning for fine-grained image classification," *IEEE Access*, vol. 13, pp. 198 054–198 064, 2025, Journal: Q1, Impact Factor: 3.6. DOI: https://doi.org/10.1109/ACCESS.2025.3635173.

4. M. Al Amin, N. Reza, and H. Y. Jung, "Lightweight network for spoof fingerprint detection by attention-aggregated receptive field-wise feature," *Electronics*, vol. 14, no. 9, p. 1823, 2025, Journal: Q2, Impact Factor: 2.6.

5. M. Al Amin et al., "High throughput biological sequence analysis using machine learning-based integrative pipeline for extracting functional annotation and visualization," *F1000Research*, vol. 13, p. 161, 2024, From SJR: Journal: Q1, Impact Factor: 0.53.

6. M. N. Yeasmin, M. Al Amin, T. J. Joti, Z. Aung, and M. A. Azim, "Advances of ai in image-based computer-aided diagnosis: A review," *Array*, vol. 23, p. 100 357, 2024, Journal: Q1, Impact Factor: 4.5.

7. M. S. Hossain et al., "Gender differences-based bioinformatics analysis to identify hub genes and key pathways in type 2 diabetes," *Informatics in Medicine Unlocked*, vol. 40, p. 101 302, 2023, Journal: Q2, Impact Factor: 6.3.

8. M. A. Rahman, M. Al Amin, M. N. Yeasmin, and M. Z. Islam, "Molecular biomarker identification using a network-based bioinformatics approach that links covid-19 with smoking," *Bioinformatics and Biology Insights*, vol. 17, p. 11 779 322 231 186 481, 2023, Journal: Q2, Impact Factor: 2.4.

9. M. S. Sarkar, M. M. Mia, M. Al Amin, M. S. Hossain, and M. Z. Islam, "Bioinformatics and network biology approach to identifying type 2 diabetes genes and pathways that influence the progression of breast cancer," *Heliyon*, vol. 9, no. 5, 2023, Journal: Q1, Impact Factor: 3.6.

10. M. Snigdha, A. Akter, M. Al Amin, and M. Z. Islam, "Bioinformatics approach to analyse covid-19 biomarkers accountable for generation of intracranial aneurysm in covid-19 patients," *Informatics in Medicine Unlocked*, vol. 39, p. 101 247, 2023, Journal: Q2, Impact Factor: 6.3.

11. M. K. Islam et al., "Bioinformatics strategies to identify shared molecular biomarkers that link ischemic stroke and moyamoya disease with glioblastoma," *Pharmaceutics*, vol. 14, no. 8, p. 1573, 2022, Journal: Q1, Impact Factor: 5.5.

12. A. S. M. Ripon Rouf et al., "Statistical bioinformatics to uncover the underlying biological mechanisms that linked smoking with type 2 diabetes patients using transcriptomic and gwas analysis," *Molecules*, vol. 27, no. 14, p. 4390, 2022, Journal: Q1, Impact Factor: 4.6.

13. M. K. Islam et al., "A secure framework toward iomt-assisted data collection, modeling, and classification for intelligent dermatology healthcare services," *Contrast Media & Molecular Imaging*, vol. 2022, no. 1, p. 6 805 460, 2022, Journal: Q3, Impact Factor: 3.16.

Conference Proceedings

1. M. K. Islam, M. Al Amin, M. R. Islam, M. N. I. Mahbub, M. I. H. Showrov, and C. Kaushal, "Spam-detection with comparative analysis and spamming words extractions," in *2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, IEEE, 2021, pp. 1–9.

2. M. K. Islam, M. N. Yeasmin, C. Kaushal, M. Al Amin, M. R. Islam, and M. I. H. Showrov, "Comparative analysis of steering angle prediction for automated object using deep neural network," in *2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, IEEE, 2021, pp. 1–7.

3. C. Kaushal, M. A. R. Refat, M. Al Amin, and M. K. Islam, "Comparative micro blogging news analysis on the covid-19 pandemic scenario," in *Proceedings of International Conference on Data Science and Applications: ICDSA 2021, Volume 2*, Springer, 2021, pp. 377–391.

4. N. I. Mahbub et al., "Sentiment analysis of microblogging dataset on coronavirus pandemic," in *2021 5th International Conference on Electrical Information and Communication Technology (EICT)*, IEEE, 2021, pp. 1–6.

5. M. A. R. Refat, M. Al Amin, C. Kaushal, M. N. Yeasmin, and M. K. Islam, "A comparative analysis of early stage diabetes prediction using machine learning and deep learning approach," in *2021 6th International Conference on Signal Processing, Computing and Control (ISPCC)*, IEEE, 2021, pp. 654–659.

Books and Chapters

1. K. Islam et al., "Iot and deep learning-based smart healthcare with an integrated security system to detect various skin lesions," in *Artificial Intelligence for Disease Diagnosis and Prognosis in Smart Healthcare*, CRC Press, 2023, pp. 219–242.

2. C. Kaushal, M. K. Islam, A. Singla, and M. Al Amin, "An iomt-based smart remote monitoring system for healthcare," in *IoT-Enabled Smart Healthcare Systems, Services and Applications*, Wiley Online Library, 2022, pp. 177–198.

Skills

Programming Languages

- ◇ Proficient in Python.
- ◇ Programming problem-solving experience on online platforms using C, C++, and Python: 2 years on UVa Online Judge (Alamin_PU) and HackerRank (Alamin_PU).

Bioinformatics and Visualization Tools

- ◇ Familiar with tools such as STRING and NetworkAnalyst for analyzing Protein-Protein Interaction, Transcription Factor Interaction, Drug-Protein Interaction, Gene-miRNA Interaction, and Protein-Disease Association.
- ◇ Experience using Enrichr for Pathways and Gene Ontology analysis.
- ◇ Proficient with Cytoscape for identifying and visualizing hub proteins.
- ◇ Skilled in using Inkscape for data visualization and graphic design.

Artificial Intelligence

- ◇ Familiar with ML models like Linear Regression and Logistic Regression.
- ◇ Understanding of frameworks such as Keras, PyTorch, and Scikit-learn.
- ◇ Comfortable with Neural Networks and Convolutional Neural Networks.
- ◇ Deep understanding of optimization, backpropagation, gradient vanishing, and activation functions like ReLU and hyperbolic tangent, and their usage.

Data Analysis

- ◇ Skilled in image processing techniques such as masking, padding, transformation, and normalization using Torch and the Python Imaging Library.
- ◇ Experienced in data cleaning, normalization, transformation, and handling missing data.
- ◇ Proficient in using libraries such as Pandas and NumPy.
- ◇ Capable of visualizing data using libraries like Matplotlib and Seaborn.

Language Proficiency

IELTS — **Overall Band Score:** 6.5

Listening: 6.5
Reading: 6.0
Writing: 6.5
Speaking: 6.0

Miscellaneous Experience

Awards and Certifications

2023 — **Scholarship:** Received a fully funded scholarship from the professor for the Master's in Computer Engineering.

2020 — **Programming Contest (Runner-up)**, Dept. of CSE, Prime University, Bangladesh.

Python Basic and Python Intermediate, HackerRank.

COVID-19: Operational Planning Guidelines and COVID-19 Partners Platform to support country preparedness and response, World Health Organization (WHO).

4th bdSIG (Bangladesh School of Internet Governance) Fellowships

References

Dr Abdullahi Chowdhury
Lecturer (Online Course Facilitator)
University of South Australia Online
University of South Australia, Australia
E-mail: abdullahi.chowdhury@unisa.edu.au

Dr. Muhammad Yeasir Arafat
Senior Postdoctoral Fellow
School of Computer Science and IT
University College Cork, Ireland
Former Assistant Professor, Chosun University, South Korea
E-mail: marafat@ucc.ie

Dr. Md Zahidul Islam
Professor
Dept. of ICT, Islamic University, Kushtia
Bangladesh
Phone: +8801733588677
E-mail: zahidimage@gmail.com

Omlan Jyoti Mondal
Lecturer
Dept. of Computer Science & Engineering
Prime University, Dhaka, Bangladesh
Phone: +8801533357058
E-mail: ojmondal@primeuniversity.edu.bd